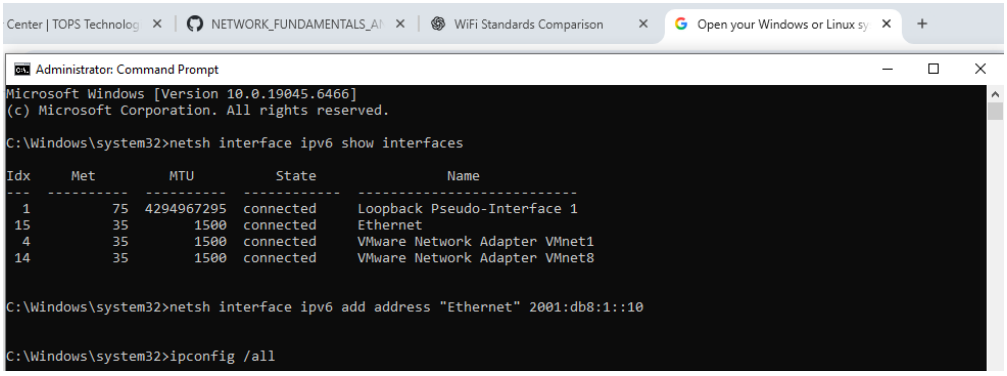
1. Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

Step 1: Open Command Prompt

Open Command Prompt as Administrator.

Step 2: Find Your Network Adapter

Type the following command: netsh interface ipv6 show interfaces



Identify your primary network adapter (e.g., Wi-Fi or Ethernet).

Step 3: Assign a Static IPv6 Address

For a Wi-Fi adapter, use:

netsh interface ipv6 add address "Wi-Fi" 2001:db8:1::10

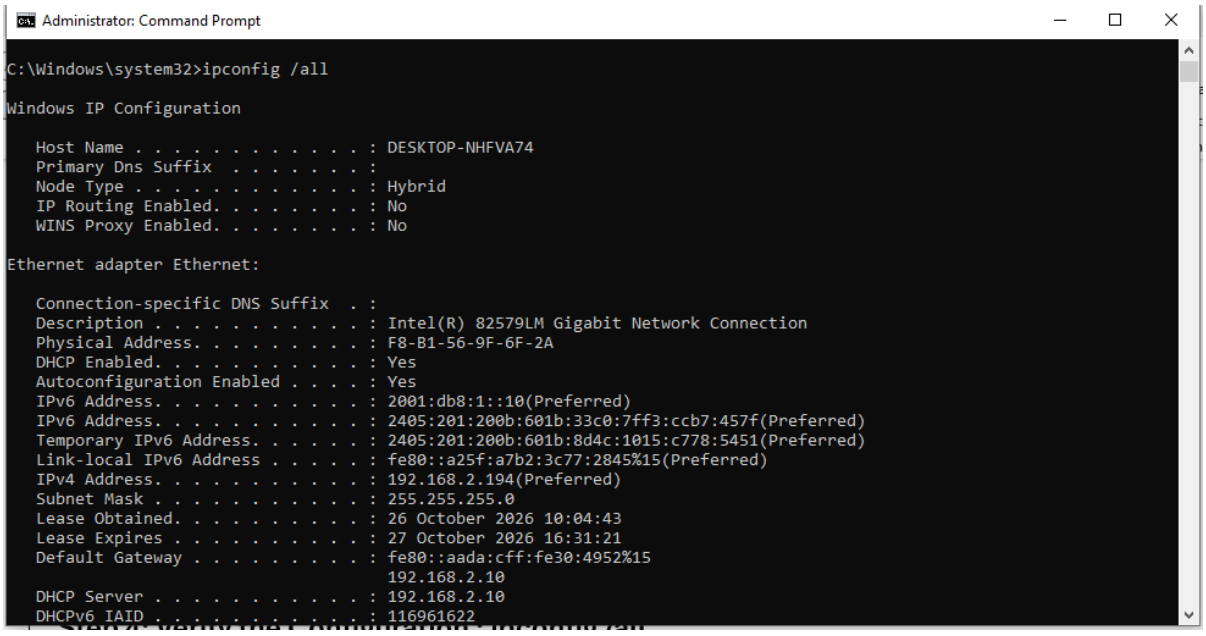
here we are replacing Wifi name as we are using Ethernet

so the command will be as follows:

netsh interface ipv6 add address "Ethernet" 2001:db8:1::10

this command has been shown in the above given figure.

Step 4: Verify the Configuration : ipconfig /all



2. Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.

Hint: Use the ping6 or ping -6 command for IPv6 testing.

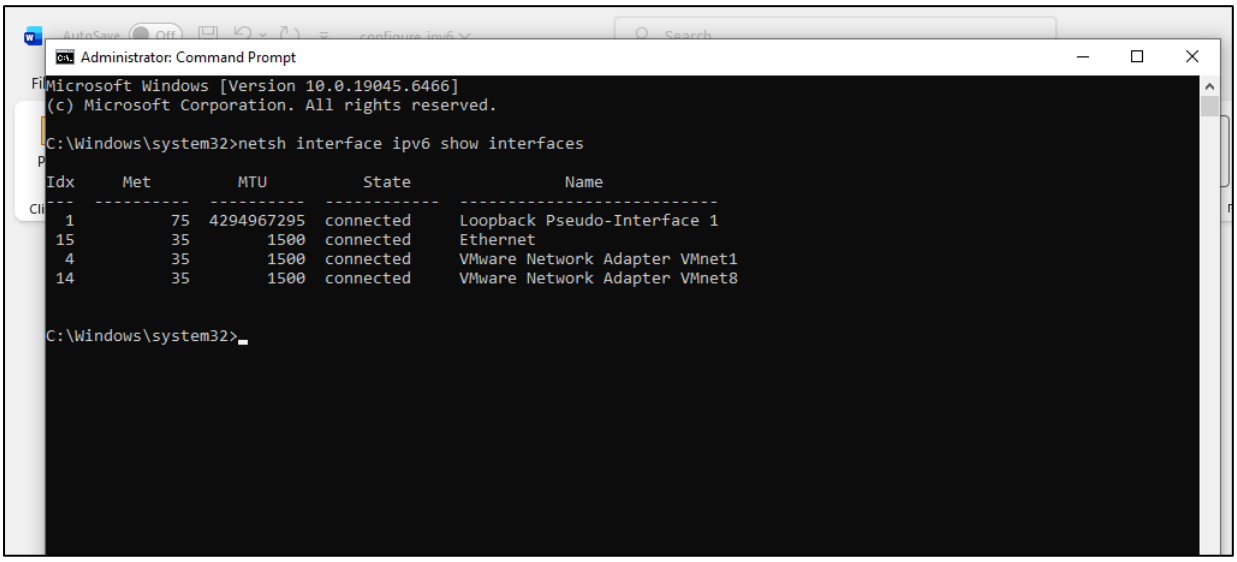
Configure a Unique Local IPv6 Address (Windows)

Step 1: Open Command Prompt

Open Command Prompt as Administrator.

Step 2: Check Network Adapter

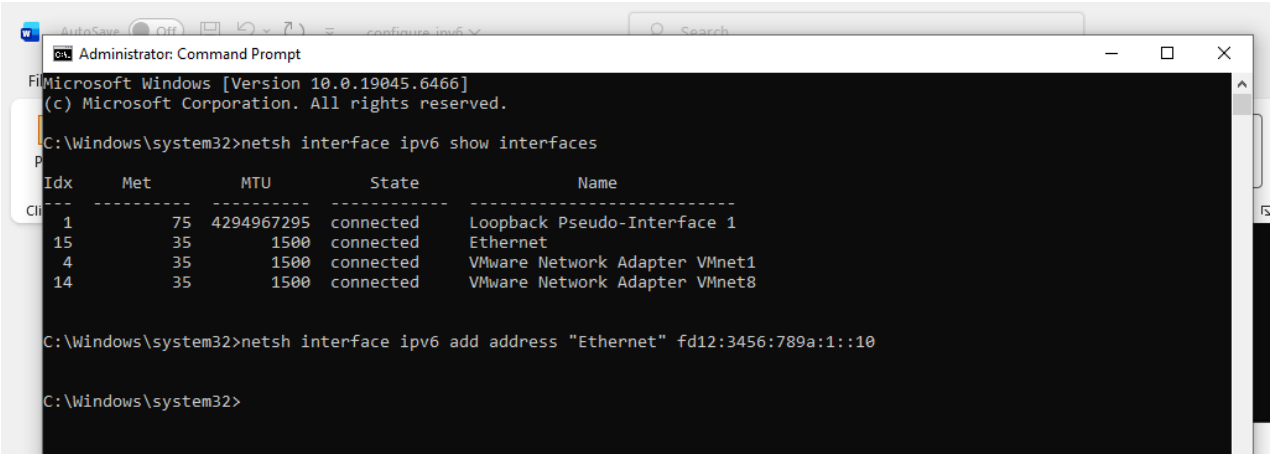
netsh interface ipv6 show interfaces



Identify your network adapter, such as Wi-Fi.

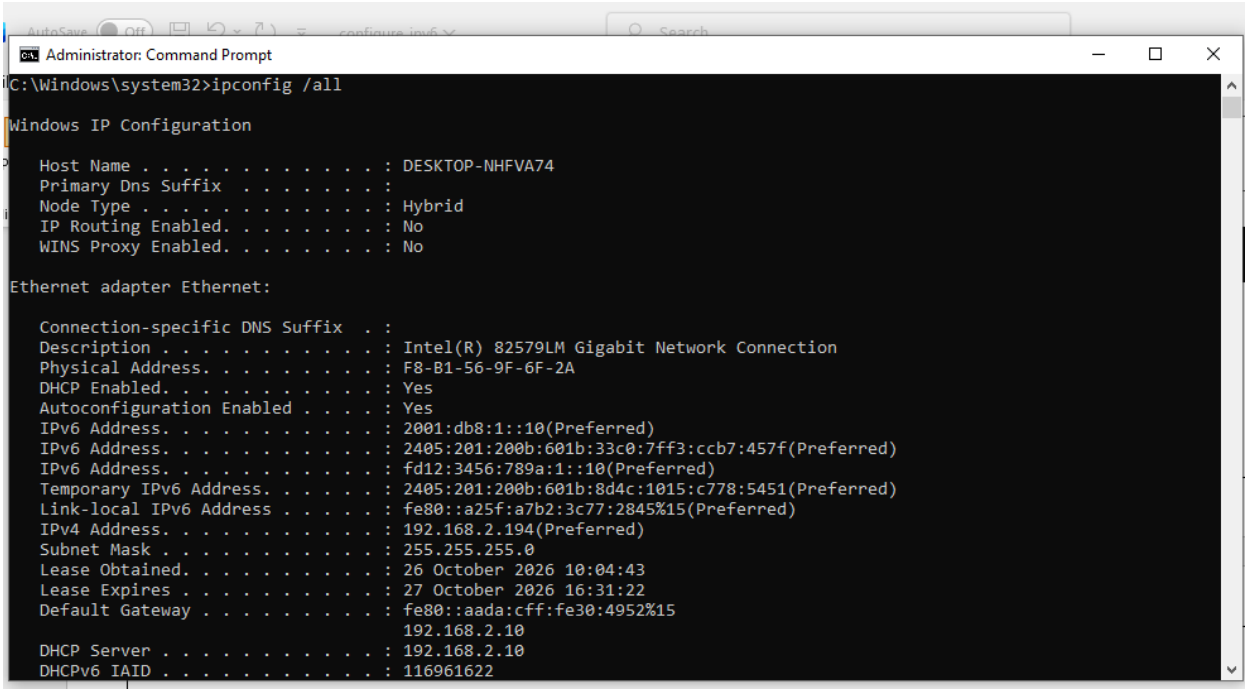
Step 3: Assign a Unique Local IPv6 Address (ULA)

netsh interface ipv6 add address "Wi-Fi" fd12:3456:789a:1::10



Step 4: Verify IPv6 Configuration

ipconfig /all



Step 5: Test Connectivity

Ensure another device on the same LAN has a different IPv6 address within the same subnet, such as fd12:3456:789a:1::20/64.

ping -6 fd12:3456:789a:1::20

expected output

```
Plain text
Reply from fd12:3456:789a:1::20: time<1ms
Reply from fd12:3456:789a:1::20: time<1ms
Reply from fd12:3456:789a:1::20: time<1ms
Reply from fd12:3456:789a:1::20: time<1ms
```

3. Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.

Constraint: Document the steps and the observed address behaviour in a text file.

Experiment: Temporary IPv6 Address Configuration

Objective: Enable a temporary IPv6 address, verify its configuration, and check whether it changes after restarting the computer or reconnecting to the network.

Steps for Windows

Step 1: Open Command Prompt as Administrator.

Step 2: Enable IPv6 Privacy Extensions.

netsh interface ipv6 set privacy state=enabled

This enables temporary IPv6 address generation for eligible IPv6 networks.

Step 3: Check IPv6 Addresses.

ipconfig /all

Look for a **Temporary IPv6 Address** under your active network adapter.

Step 4: Record the Address.

Copy the temporary IPv6 address before restarting or disconnecting.

Step 5: Disconnect and Reconnect Wi-Fi.

Reconnect to the same network and run:

ipconfig /all

Compare the temporary IPv6 address with the previously recorded address.

Step 6: Restart the Computer.

After restarting, run ipconfig /all again and compare the addresses.

Expected Observation

Temporary IPv6 addresses are designed to improve privacy by reducing long-term address tracking. They may change after a network reconnection or reboot, but **a change is not guaranteed immediately**.

Temporary global IPv6 address generation also depends on the network providing suitable IPv6 configuration.

4. Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

Zomato-Style Restaurant Search Using Python

```
# Restaurant names and IPv6 addresses
restaurants = {
   "Pizza Hut": "2001:db8::1",
   "Dominos": "2001:db8::2",
   "Burger King": "2001:db8::3",
   "KFC": "2001:db8::4",
   "Subway": "2001:db8::5"
}
```

```
# Take input from user
search = input("Enter restaurant name: ").strip().lower()

# Search for matching restaurants
found = False

for name, ipv6 in restaurants.items():
    if search in name.lower():
        print(f"Restaurant: {name} | IPv6 Address: {ipv6}")
        found = True

if not found:
    print("Restaurant not found!")
```

Example Output

Input:

Enter restaurant name: pizza

Output:

Restaurant: Pizza Hut | IPv6 Address: 2001:db8::1

5. Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

IPv6 Subnetting Scenario

Objective: Divide an IPv6 network into at least three subnets and manually configure an IPv6 address on a Windows computer.

1. AI-Generated Subnetting Scenario

Suppose a company has three departments: HR, IT, and Sales. Each department requires a separate IPv6 subnet.

Main IPv6 Network: fd12:3456:789a::/48

Department	IPv6 Subnet	Example IPv6 Address
HR	fd12:3456:789a:1::/64	fd12:3456:789a:1::10
IT	fd12:3456:789a:2::/64	fd12:3456:789a:2::10
SALES	fd12:3456:789a:3::/64	fd12:3456:789a:3::10

Each department has a separate /64 subnet.

2. Windows Configuration Steps

Step 1: Open Command Prompt as Administrator.

Step 2: Identify your network adapter.

netsh interface ipv6 show interfaces

Step 3: Assign an IPv6 address from the IT department's subnet.

netsh interface ipv6 add address "Wi-Fi" fd12:3456:789a:2::10/64

Step 4: Verify the configuration.

ipconfig /all

Step 5: Check whether the IPv6 address appears under the Wi-Fi adapter.